Hybrid Clustering for Customer Segmentation Using RFM Analysis

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***Abstract*—Customer segmentation plays a crucial role in modern data-driven businesses by enabling personalized marketing, improved customer retention, and strategic decision-making. This study focuses on segmenting customers based on their purchasing behavior using an online retail dataset.**

The raw transactional data is first transformed into meaningful customer-level features using RFM (Recency, Frequency, Monetary) analysis, which captures how recently, how often, and how much customers spend. These features are then normalized to ensure fair comparison across different scales.

Multiple clustering techniques, including K-Means, DBSCAN, and Fuzzy C-Means (FCM), are implemented to group customers into distinct segments. In addition, hybrid clustering approaches are developed by combining these algorithms to leverage their individual strengths.

To evaluate clustering performance, a manually implemented silhouette score is used, providing insight into cluster compactness and separation. Experimental results show that K-Means performs best for structured datasets, while hybrid models offer improved robustness in handling noise and complex data distributions.

The study highlights the importance of selecting appropriate clustering techniques based on dataset characteristics and demonstrates the practical application of hybrid clustering in customer segmentation.

I. Introduction

In today's data-driven world, businesses generate large volumes of customer transaction data. Analyzing this data effectively is essential for understanding customer behavior and improving business strategies. One of the most important applications of such analysis is customer segmentation, where customers are grouped based on similar characteristics or purchasing patterns.

Customer segmentation enables organizations to perform targeted marketing, improve customer retention, and increase overall profitability. Instead of treating all customers equally, businesses can design personalized strategies for different customer groups such as high-value customers, frequent buyers, or inactive users.

Clustering is a widely used unsupervised machine learning technique for customer segmentation. Unlike supervised learning, clustering does not require labeled data and instead identifies patterns and groupings based on similarities in the data. However, different clustering algorithms have different strengths and limitations [1].

K-Means is one of the most popular clustering algorithms due to its simplicity and efficiency [2], but it assumes that clusters are spherical and is sensitive to noise. DBSCAN, a density-based algorithm, can detect outliers and handle arbitrarily shaped clusters but depends heavily on parameter selection [1]. Fuzzy C-Means (FCM) allows data points to belong to multiple clusters, making it suitable for overlapping data, but it often results in less clear cluster boundaries [1].

To overcome the limitations of individual algorithms, hybrid clustering approaches are introduced. Hybrid models combine multiple clustering techniques to leverage their strengths, such as combining the stability of K-Means, the noise-handling capability of DBSCAN, and the flexibility of FCM.

In this study, customer segmentation is performed using an online retail dataset. The raw transactional data is first transformed into meaningful customer-level features using RFM (Recency, Frequency, Monetary) analysis. These features are then normalized and used as input for clustering algorithms.

Multiple clustering techniques, including K-Means, DBSCAN, Fuzzy C-Means, and hybrid combinations of these methods, are implemented and compared. The performance of each model is evaluated using a manually implemented silhouette score, which measures the quality of clustering based on cohesion and separation.

The main objective of this work is to analyze and compare the effectiveness of individual and hybrid clustering approaches and to understand under which conditions hybrid models provide better results. This study highlights that the choice of clustering technique should depend on the characteristics of the dataset rather than assuming one method is universally superior.

II. Dataset Description

The dataset used in this study is an online retail transaction dataset [3], where each record represents a single purchase made by a customer. It contains detailed information about customer transactions, including:

- **CustomerID:** Unique identifier for each customer
- **InvoiceNo:** Unique transaction identifier
- **InvoiceDate:** Date and time of purchase
- **Quantity:** Number of items purchased
- **UnitPrice:** Price per item

This dataset is transactional in nature, meaning it contains multiple records for the same customer across different purchases. Therefore, it needs to be transformed into a customer-level dataset for effective analysis.

Objective: The main objective is to segment customers into meaningful groups based on their purchasing behavior. This helps businesses identify:

- High-value customers
- Frequent buyers
- Inactive or low-engagement customers

Such segmentation is useful for targeted marketing, customer retention strategies, and business decision-making.

III. Data Preprocessing

Raw data often contains inconsistencies, missing values, and invalid entries, which can negatively affect clustering performance. Therefore, several preprocessing steps were applied to clean and prepare the dataset.

Steps performed:

- **Handling Missing Values:** Rows containing missing values were removed to ensure data integrity and avoid errors during processing.
- **Filtering Invalid Transactions:** Transactions with negative or zero values in Quantity and UnitPrice were removed, as they represent returns or incorrect entries.
- **Feature Creation:** A new feature, *TotalAmount*, was created using:

$$\text{TotalAmount} = \text{Quantity} \times \text{UnitPrice}$$

  This represents the total value of each transaction.
- **Date Conversion:** The InvoiceDate column was converted into datetime format to enable time-based calculations such as recency.

Importance of Preprocessing:

- Ensures accuracy and reliability of clustering results
- Removes noise and inconsistencies from the dataset
- Prepares the data for meaningful feature extraction (RFM)

IV. RFM Feature Engineering

RFM is a widely used technique in customer analytics that transforms transactional data into meaningful customer-level features [4]. It consists of three components:

- **Recency (R):** Measures how recently a customer made a purchase. Customers who purchased recently are more likely to return.
- **Frequency (F):** Represents how often a customer makes purchases. Higher frequency indicates loyal customers.
- **Monetary (M):** Indicates how much money a customer spends. High monetary value customers are more profitable.

Why RFM is used:

- Converts raw transactional data into structured numerical features
- Captures customer behavior effectively
- Helps distinguish between high-value, inactive, and regular customers

Implementation in this project:

- Recency = difference between latest date and last purchase
- Frequency = count of transactions per customer
- Monetary = total spending of each customer

V. Normalization

$$X = \frac{X - \mu}{\sigma} \tag{1}$$

Purpose:

- Ensures all features have equal importance
- Prevents Monetary from dominating clustering

VI. Architecture Diagram

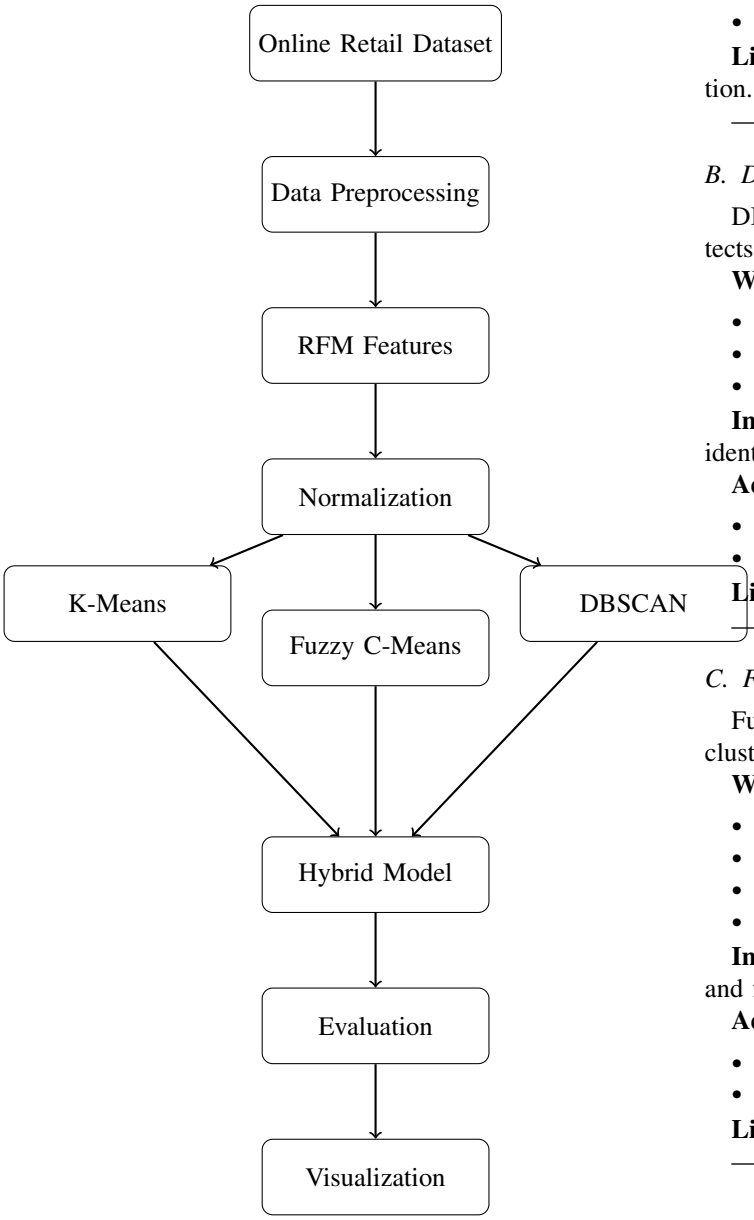

Fig. 1. Hybrid Clustering Architecture

VII. Clustering Algorithms

A. K-Means

K-Means is a centroid-based clustering algorithm that partitions data into k clusters by minimizing intra-cluster distance.

Working:

- Initialize centroids randomly
- Assign each point to nearest centroid using Euclidean distance
- Update centroids as mean of assigned points
- Repeat until convergence

In this project: Distance is calculated using numpy norms, and labels are assigned using minimum distance.

Advantages:

- Fast and efficient
- Works well for structured data

Limitations: Sensitive to noise and initial centroid selection.

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B. DBSCAN

DBSCAN is a density-based clustering algorithm that detects noise [5].

Working:

- Uses eps (radius) and min_samples
- Dense regions form clusters
- Sparse points are labeled as noise (-1)

In this project: Distance is computed and neighbors are identified using threshold.

Advantages:

- Detects outliers automatically
- Works with arbitrary shapes

Limitations: Sensitive to parameter selection.

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C. Fuzzy C-Means (FCM)

Fuzzy C-Means allows data points to belong to multiple clusters [6].

Working:

- Initialize membership matrix
- Compute cluster centers using weighted average
- Update membership values based on distance
- Repeat until convergence

In this project: Membership matrix is updated iteratively and final cluster is chosen using maximum membership.

Advantages:

- Handles overlapping clusters
- More flexible than K-Means

Limitations: Produces less clear cluster boundaries.

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VIII. Hybrid Models

Hybrid clustering combines multiple algorithms to improve results.

A. KMeans + FCM

- Voting-based approach
- If both agree $\rightarrow$ assign same cluster
- Else $\rightarrow$ choose KMeans

B. FCM + DBSCAN

- DBSCAN removes noise
- FCM clusters remaining points

C. KMeans + DBSCAN

- DBSCAN filters outliers
- KMeans clusters clean data

D. Combined Hybrid

- DBSCAN removes noise
- KMeans and FCM combined using voting

IX. Evaluation

Clustering performance is evaluated using Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \tag{2}$$

Where:

- $a(i)$ = intra-cluster distance
- $b(i)$ = nearest cluster distance

Interpretation:

- Close to 1 $\rightarrow$ good clustering
- Close to 0 $\rightarrow$ overlapping clusters
- Negative $\rightarrow$ poor clustering

In this project: Silhouette score is manually implemented to understand clustering quality without using libraries.

X. Results

TABLE I
Silhouette Scores

Model	Score
K-Means	0.6038
DBSCAN	0.2993
FCM	-0.0474
KM+FCM	0.4392
FCM+DBSCAN	0.1340
KM+DBSCAN	0.1803
Hybrid ALL	0.1401

XI. Visualization and Analysis

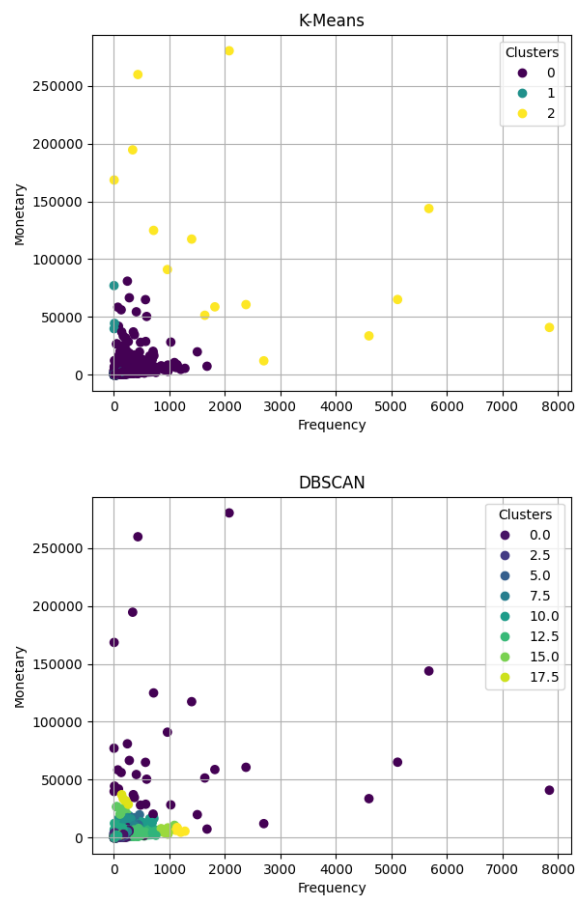

Fig. 2. K-Means (clear clusters) vs DBSCAN (noise detection)

Observation: K-Means forms well-separated clusters, while DBSCAN detects outliers.

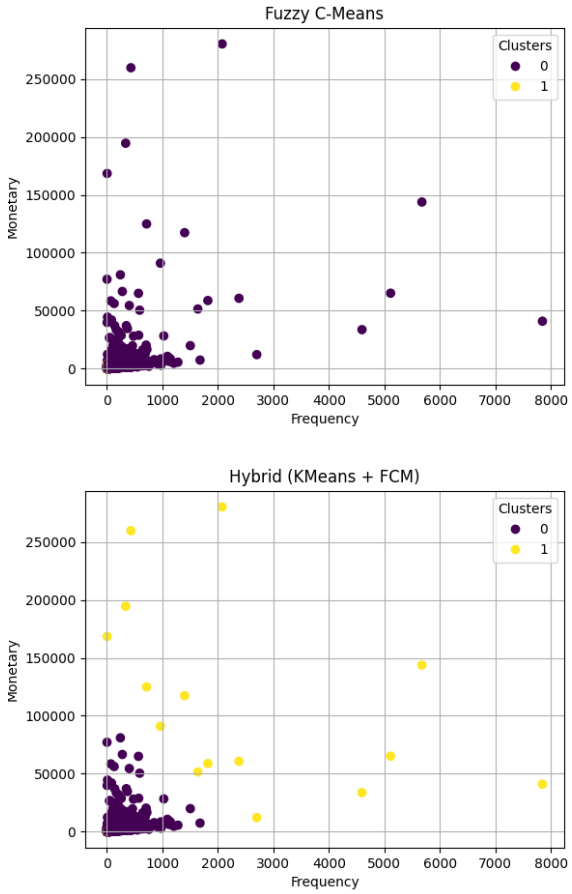

Fig. 3. FCM vs KMeans+FCM

Observation: FCM produces overlapping clusters; hybrid improves structure.

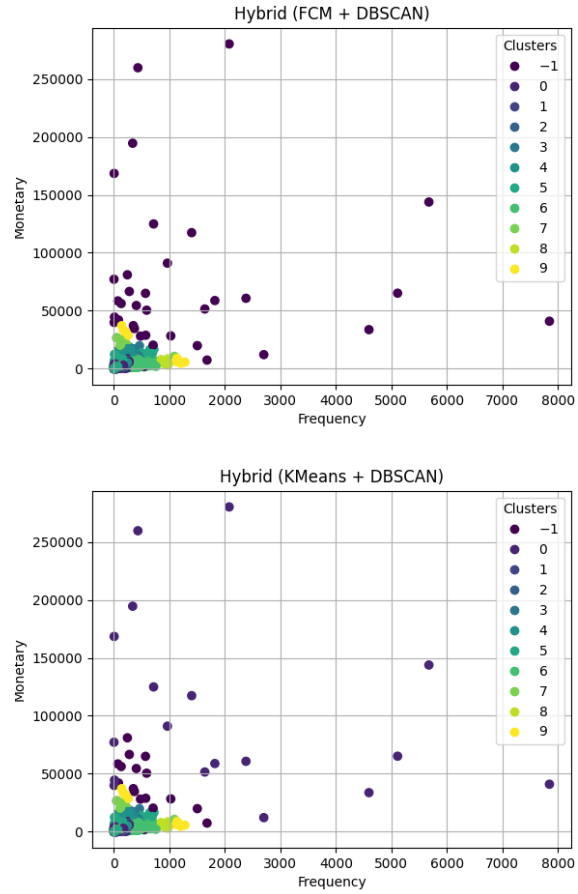

Fig. 4. FCM+DBSCAN vs KMeans+DBSCAN

Observation: Noise filtering improves robustness but reduces cluster clarity.

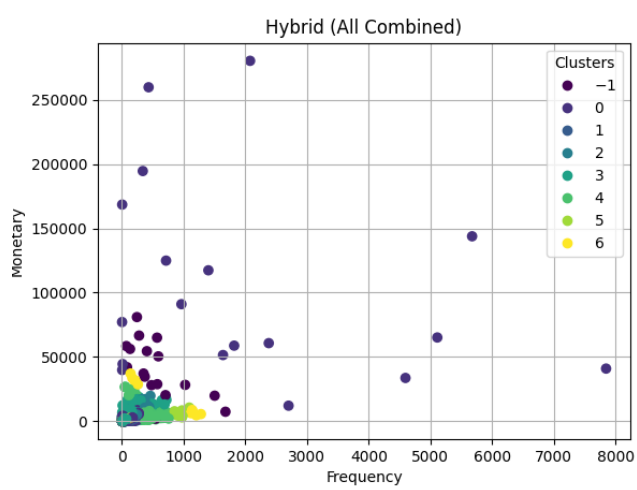

Fig. 5. Combined Hybrid Model

Observation: Combined hybrid introduces complexity and reduces separation.

XII. Discussion

The experimental results highlight important insights about clustering behavior on the RFM-based dataset.

K-Means achieved the highest silhouette score (0.6038), indicating strong cluster separation and compactness. This suggests that the dataset has relatively well-defined and structured clusters, where distance-based partitioning works effectively.

DBSCAN performed moderately with a silhouette score of 0.2993. It successfully identified noise points (outliers), which improves robustness. However, the removal of data points reduces cluster density, resulting in lower overall performance.

Fuzzy C-Means (FCM) showed poor performance with a negative silhouette score. This is due to its soft clustering nature, where data points belong to multiple clusters, leading to overlapping boundaries and reduced clarity in segmentation.

Hybrid models showed mixed results:

- KMeans + FCM improved over FCM but did not outperform K-Means
- FCM + DBSCAN suffered due to combining weak clustering with noise filtering
- KMeans + DBSCAN reduced performance due to loss of data after noise removal
- Combined Hybrid introduced excessive complexity, leading to unstable clusters

Overall, the results indicate that hybrid models improve robustness but may reduce cluster clarity when applied to well-structured datasets.

XIII. When Hybrid Clustering is More Effective

Hybrid clustering is not always superior to individual algorithms, but it becomes highly effective under certain conditions.

Hybrid clustering performs better when:

- The dataset contains significant noise or outliers
- Clusters have irregular shapes or varying densities
- Data points overlap significantly
- Single algorithms fail to capture complex patterns

Examples:

- DBSCAN can remove noise, after which K-Means can form better clusters
- FCM can handle overlapping data where K-Means fails
- Combined hybrid methods can balance flexibility and stability

However, hybrid clustering may not perform well when:

- The dataset is already clean and well-structured
- Clusters are clearly separable
- Additional complexity introduces instability

In this study, the dataset is well-structured due to RFM transformation, which explains why K-Means outperformed hybrid approaches.

XIV. Conclusion

This study presented a comprehensive analysis of customer segmentation using RFM (Recency, Frequency, Monetary) feature engineering combined with multiple clustering techniques, including K-Means, DBSCAN, Fuzzy C-Means (FCM), and their hybrid combinations.

The transformation of raw transactional data into RFM features played a crucial role in structuring the dataset, enabling effective clustering. By converting customer behavior into numerical representations, RFM provided a meaningful foundation for identifying patterns in purchasing behavior.

Among the individual clustering algorithms, K-Means achieved the highest silhouette score, indicating strong cluster cohesion and separation. This result suggests that the dataset, after preprocessing and normalization, exhibits relatively well-defined and compact clusters, making centroid-based clustering highly effective.

DBSCAN demonstrated its strength in identifying noise and outliers, which improves robustness in real-world scenarios. However, the removal of noise points led to reduced cluster density, affecting the overall silhouette score. Fuzzy C-Means, on the other hand, introduced flexibility by allowing data points to belong to multiple clusters. While this is beneficial for overlapping data, it resulted in less distinct cluster boundaries and lower evaluation performance.

Hybrid clustering approaches were implemented to combine the strengths of individual algorithms. Techniques such as KMeans+FCM, FCM+DBSCAN, and KMeans+DBSCAN aimed to improve clustering robustness and adaptability. The combined hybrid model further attempted to integrate all approaches. Although hybrid models provided better handling of noise and overlapping data, they did not outperform K-Means in this study. This is primarily because the dataset was already well-structured after RFM transformation, reducing the need for complex hybrid strategies.

The results clearly highlight that there is no universally best clustering algorithm. The effectiveness of a clustering technique depends heavily on the nature of the dataset. For clean, structured, and well-separated data, simpler algorithms like K-Means are more efficient and effective. In contrast, hybrid models become more useful in scenarios involving noisy, complex, or overlapping datasets.

In conclusion, this study emphasizes the importance of understanding data characteristics before selecting a clustering approach. Future improvements can explore advanced ensemble clustering methods, parameter optimization techniques, and the application of deep learning-based clustering models to further enhance segmentation performance in more complex real-world datasets. The implementation of this work is publicly available online [7].

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